

Architectural guidance

This section attempts to answer one key question: which tool should be used for which use case and how do we decide the best option?

As evidenced in the previous sections, there are a plethora of options available for establishing connectivity to External Data Sources and Ingesting data into the Data Lake's Intake Tier.

Choosing a tool for ingestion depends primarily on the use case you are attempting to implement using the Data Lake. Many implementations of Data Lake end up using multiple tools together to acquire and process data. We also see that the market is flooded with umpteen numbers of tools that make decision making very difficult.

In this section, we will try to provide a crisp overview of the things to be considered while selecting a tool. This is by no means an exhaustive coverage, but it tries to give you enough depth so that you will be in a decent position to extrapolate using this knowledge and make better decisions.

The choice of the tool invariably starts with the now ever-obvious use cases, which can be broadly categorized into two major groups based on the type of data to be ingested.

Structured data use cases

All the Data Lake use cases that derive insights by integrating with operational systems or enterprise data warehouses, to perform traditional business intelligence, feed OLAP applications and generate basic visualization such as reports and drilldowns that ingest Structured Data into the Data Lake's Intake Tier. The ingestion of structured data can be scheduled to be performed at regular intervals. Most of this data is *Pulled* from

External Data Sources. The tools available to ingest data for this category of use cases are Sqoop and WebHDFS, for ingesting delimited files, Splunk (to an extent), and so on.

Semi-structured and unstructured data use cases

These are advanced use cases of implementing a Data Lake to wring the last drop of value from the data assets within and without the organization. For instance, if the businesses' use case need analytic insights from the following:

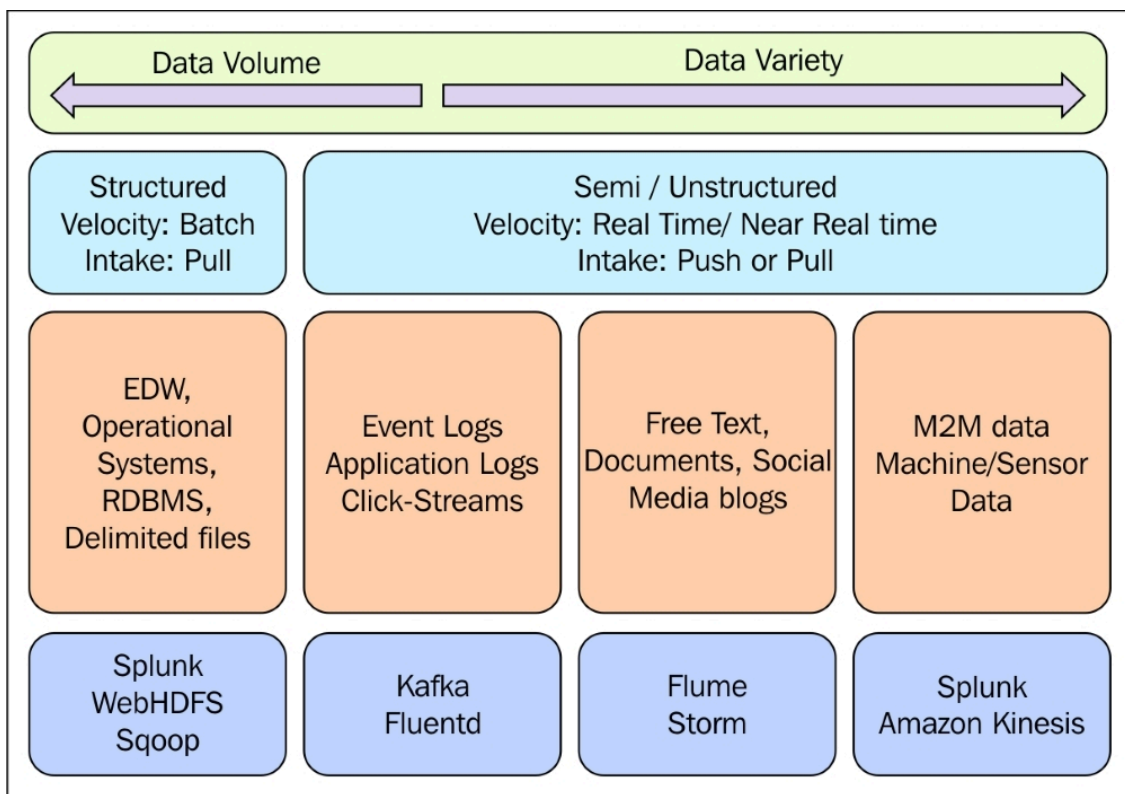
- Performing social media analysis
- Building recommendation engines for e-commerce sites
- Advanced visualization techniques such as Heat Map and Tag Cloud
- Analyzing In-memory event streams
- Acquiring sensor data and detecting anomalies in real time
- Conducting simulations on massive behavioral data

In all of the preceding business use cases, you were dealing with mostly unstructured data that has to be ingested into the Data Lake as soon as possible to generate insights. These use cases typically ingest on the fly data such as sensor exhaust, social chatter, clickstreams, and so on.

There are a few other use cases such as natural language processing of voluminous text, for which the speed of ingest for unstructured data doesn't really matter.

The tools available to ingest data for these categories of use cases are Flume, Kafka Storm, Fluentd, and Splunk, and so on.

The following figure simplifies this decision-making process by taking into consideration the preceding combinations and suggests an indicative list of tools that can be used for each use case:



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This list of Big Data tools and technologies can then be evaluated to see whether they fit in for the specific use case requirement in your organization's context. The next section details out each of the tools.

Big Data tools and technologies

Now that we have understood the various aspects of the Data Intake tier, the following sections provide an overview of Big Data tools for ingesting structured, semi-structured, and unstructured data as discussed in the preceding section.

Ingestion of structured data

Let us now explore a few Big Data technologies used for ingesting structured data from relational databases and structured delimited files into the Data Lake.

Sqoop

Sqoop, a Hadoop ecosystem component, allows you to efficiently transfer peta-byte scale data between RDBMS and Hadoop. This tool helps to acquire the data from RDBMS, by efficiently interacting with traditional

RDBMS servers to perform the import and export of data. Loading petabyte scale data into the Data Lake from production databases is a challenging task as it involves the overhead of handling issues such as data consistency, failure handling, and parallelism, and this is where Sqoop excels.

With Sqoop, you can perform full data extraction, bulk data extraction, CDC, and online or offline extraction. Sqoop allows you to select particular columns or filter particular rows from a table to load it into the Data Intake Tier. It also supports CDC, that is able to retrieve only the rows that were not present in the set of rows that were previously imported.

Use case scenarios for Sqoop

The scenarios in which Sqoop can be used to ingest data are as follows:

- Ingest structured data from relational databases into Hadoop
- Ingest mainframe datasets into Hadoop
- Perform full data extraction, CDC, bulk data extraction, and online or offline extraction
- Extract a subset of data by selecting specific columns or filter rows to match a condition

TIP

A compact data format such as Avro can be used in combination with compression when you are importing massive amounts of data. Compression codecs can be passed as arguments to compress the files according to the codec of your choice.

NOTE

More information on Sqoop can be found at the following URL:

<http://sqoop.apache.org/docs/1.4.6/index.html>

In Data Lake, there are external upstream systems that use a *Push*-based approach to ingest data into HDFS. To ingest the data without the need for Hadoop to be installed on these systems, they will need access to the HDFS and the data stored in it; WebHDFS was developed to address this need.

WebHDFS allows external clients to access Hadoop in a language-agnostic way; it retains native Hadoop protocol security and inherent parallelism that Hadoop offers. All the HDFS operations are supported by WebHDFS.

Use case scenarios for WebHDFS

WebHDFS can be used in use cases where there is a need to do the following:

- Ingestion of structured as well as unstructured files into HDFS in a batch or streaming mode, without the need for Hadoop to be installed on the client
- Steady addition of data into log files.

NOTE

More information on WebHDFS can be found at the following URL:

<https://hadoop.apache.org/docs/r1.0.4/webhdfs.html>

Ingestion of streaming data

The following subsections provide an overview of a few Big Data technologies used for ingesting large-scale streaming data.

Apache Flume

Apache Flume addresses the need for collecting, aggregating, and ingesting large-scale streaming data into the Data Lake's Intake Tier. Flume was designed to be distributed; it is robust, fault-tolerant, and reliable.

Flume collects streaming data in a centralized store, primarily Hadoop. This streaming data can be utilized to perform real-time analysis or batch analysis depending upon the use case.

Use case scenarios for Flume

Flume can be used in use cases where there is a need to ingest real-time or micro batch data such as application logs, network protocols, IoT devices, and social media data into the Data Lake's Intake Tier

NOTE

More information on Flume can be found at the following URL:

<https://flume.apache.org/releases/1.6.0.html>

Fluentd

Fluentd was developed to address the collection and transport aspects of centralized logging; it provides a unified logging layer that abstracts data sources from backend systems.

Fluentd supports file and memory-based buffering to prevent data loss. It has a plugin system that allows easy extensibility to read from multiple sources and write to multiple sinks.

Use case scenarios for Fluentd

Fluentd can be used in use cases where there is a need to ingest real-time or micro batch data such as application logs, network protocols, IoT devices, and social media data into the Data Lake's Intake Tier.

NOTE

More information on Fluentd can be found at the following URL:

<http://docs.fluentd.org/articles/quickstart>

Kafka

Kafka was built as a unified platform to address the need to handle real-time data feeds and integrate these feeds with external data sources. Kafka provides scalable, fast, and persistent messaging that enables parallel data loads into the Data Lake.

Huge amounts of terabyte scale data can be ingested by Kafka, allowing the implementation of use cases that cannot be handled by traditional systems.

Use case scenarios for Kafka

Kafka can be used in use cases where there is a need to ingest real-time or micro batch data such as application logs, network protocols, IoT devices, and social media data into the Data Lake's Intake Tier.

NOTE

More information on Kafka can be found at the following URL:

<http://kafka.apache.org/documentation.html>

Amazon Kinesis

Kinesis provides the ability to aggregate and store terabytes of data from multiple sources such as clickstreams, social media data, logs, GPS tracking, and financial transactions data to perform real-time processing on it. Kinesis supports the consumption of the data by multiple applications that can perform various actions simultaneously and independently.

Use case scenarios for Kinesis

Kinesis can be used in use cases where there is a need to set up high-capacity data pipes to rapidly ingest real-time data such as clickstreams, social media data, logs, GPS tracking, and financial transactions data into the Data Lake's Intake Tier.

NOTE

More information on Kinesis can be found at the following URL:

<http://aws.amazon.com/kinesis/developer-resources/>

Apache Storm

Storm is an event processing system built to rapidly ingest data from event streams and is often used to acquire data into Data Lake's Hadoop storage and OLAP systems.

Storm is data type agnostic and processes any data type; any programming language can use Storm. It guarantees that stream processing happens without data loss.

Use case scenarios for Storm

Storm can be used in use cases where there is a need to rapidly ingest real-time data such as clickstreams, social media data, logs, GPS tracking, and financial transactions data into the Data Lake's Intake Tier.

NOTE

More information on Storm can be found at the following URL:

<https://storm.apache.org/documentation/Home.html>
